IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

SPEECH PROCESSING

Friday, 17 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P. Naylor
Second Marker(s) : W. Dai

© Imperial College London

Speech Processing

1. a) Briefly explain the relative advantages and disadvantages of ***statistical model-based*** and ***end-to-end trained*** Automatic Speech Recognition (ASR) systems. [4]

   b) Consider a statistical model-based ASR system, including a trigram language model.

      i) Draw a block diagram of this ASR system. [2]

      ii) Using the notation

      $$w[i] \text{ for } i = \ldots, -2, -1, 0, 1, 2, \ldots$$

      to represent a word in a word sequence with $i = 0$ denoting the current word, write down the equation for the output of the language model. [2]

      iii) Explain how the language model can be trained when the quantity of training data is limited. [3]

   c) Consider a hidden Markov model-based ASR system for which a speech utterance is represented by T frames of feature vectors, $\mathbf{x}_1$, $\mathbf{x}_2$, ..., $\mathbf{x}_T$. The utterance is compared with a Hidden Markov model having S states. The transition probability from state i to state j of the model is denoted by a_{ij} and the output probability density of frame t in state i is denoted by $d_i(\mathbf{x}_t)$.

      The highest probability density that the model generates frames

      $$\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_t$$

      from any sequence of states for which frame 1 is in state 1 and frame t is in state s is denoted $B(t,s)$.

      i) Explain how, for $t > 1$, $B(t,s)$ can be expressed in terms of $B(t-1,i)$ for $i = 1,2,\ldots,S$. Indicate the values that should be given to $B(1,i)$. [3]

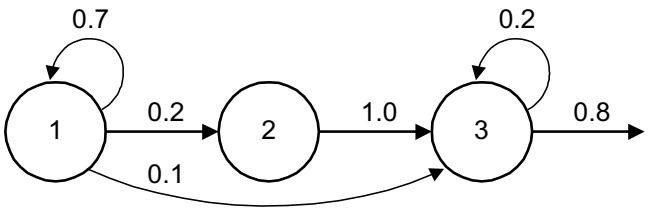
      ii) In a speech recognition experiment, an utterance represented by $T = 5$ frames is compared with a hidden Markov model that has 3 states. Table 1 shows the output probability density of each frame from each state of the model and Figure 1.1 shows the state diagram of the model. The figures in circles are the state numbers and the figures on arrows indicate the transition probabilities.

      Determine the value of $B(5,3)$ and the state sequence to which it corresponds. Use six decimal places in your calculations. [6]

	frame				
	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$
state 1	0.5	0.1	0.1	0.6	0.3
state 2	0.2	0.1	0.2	0.2	0.2
state 3	0.3	0.4	0.5	0.4	0.5

Table 1

Figure 1.1



2. a) Consider the simplified block diagram of a speech coder shown in Fig. 2.1.

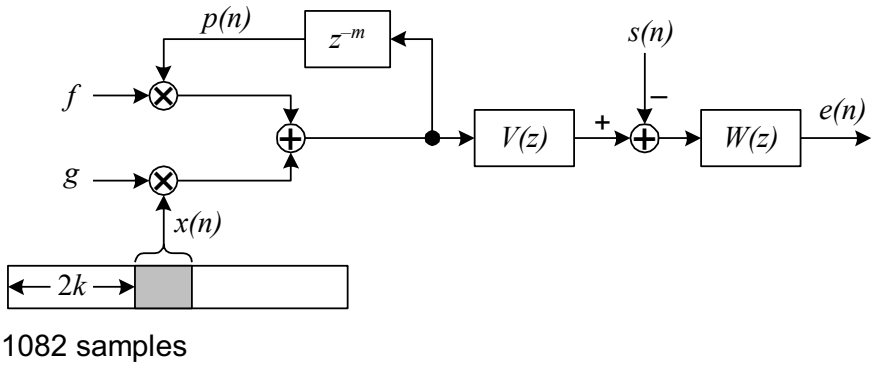


Figure 2.1: Simplified block diagram of a speech coder

i) Describe the meaning and purpose of the signals labelled $p(n)$, $x(n)$, $s(n)$ and $e(n)$. [4]

ii) Identify which elements of this coder are used to represent the periodic components of the speech excitation. [2]

iii) Explain in detail how the non-periodic components of the speech excitation signal are represented in this coder and how they are used to code a speech signal during the operation of the coder. *Hint: it is sufficient to explain what operations and functions are performed and it is not necessary to elaborate mathematically on how the calculations are performed.* [3]

b) i) Consider an analogue, continuous-time speech signal $s(t)$ that is to be digitally encoded as a quantized discrete-time signal $s(n)$. Given that $p(s)$ denotes the probability density function of s, state which characteristics of $p(s)$ would make non-uniform quantization more effective than uniform quantization. Justify your answer. [3]

ii) Explain the aims of *waveform coding* and *perceptual coding*, in the context of coding speech signals. State an example of each approach and the corresponding coding gains in each case that could be typically expected. [3]

iii) A PCM-encoded speech signal $s(n)$ uses 16 bits per sample. The signal $s(n)$ is next transcoded to μ-law encoding in which S, e and m denote the sign, exponent and mantissa bits, respectively, and the quantization scheme has bin centres at

$$\pm\Big((m+16.5)2^e - 16.5\Big).$$

A particular sample of $s(n)$ has the value 66. Find the values of S, e and m in the μ-law encoding of this sample, and state the absolute value of the coding error associated with this sample. [5]

3. a) i) Let $V(z)$ denote a vocal tract filter. Give the formula for the corresponding complex cepstrum representation, and explain the advantages of the complex cepstrum for speech processing. [3]

ii) Let the unit step function be $u(n)$. Consider a signal $x(n)$ and a scalar constant a such that

$$x(n) = a^n u(n)$$

with $|a| < 1$. Find the complex cepstrum of $x(n)$. [4]

You may use the identity

$$\log(1-\alpha) = -\sum_{n=1}^{\infty} \frac{\alpha^n}{n} \text{ for } |\alpha| < 1.$$

b) Consider the mel scale mapping of frequency in Hertz given by

$$\text{mel}(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right).$$

i) Let

$$\Delta f_1 = 200 - 100 \text{ Hz},$$
$$\Delta f_2 = 2600 - 2500 \text{ Hz}.$$

Calculate Δm_1 and Δm_2 as the differences in mel corresponding to Δf_1 and Δf_2, respectively.

Comment on your answer. [4]

c) The Mel-frequency cepstral coefficients can be computed using (among other functions) the FFT of a frame of speech signal samples. With the aid of a block diagram, explain in detail all the steps needed. [5]

d) A frame of speech samples is represented by a feature vector, $\mathbf{x}$, containing N Mel-frequency cepstral coefficients. The feature vector elements are known to be somewhat correlated.

Let $\mathbf{z}$ denote a vector with the same dimension as $\mathbf{x}$. Each element of $\mathbf{z}$ is an independent Gaussian distributed random variable.

It is decided to model $\mathbf{x}$ using a non-singular matrix $\mathbf{A}$ as

$$\mathbf{x} = \mathbf{A}\mathbf{z}.$$

i) Write down an expression for the covariance matrix of $\mathbf{x}$ in terms of matrix $\mathbf{A}$. [1]

ii) Hence, find an expression for the probability density function of $\mathbf{x}$. [3]

4. a) Draw a labelled block diagram of a single-channel noise reduction system for speech enhancement [4]

b) A particular single-channel noise reduction system implements

amplitude spectral subtraction.

The noisy signal in time frame l and at frequency bin k is denoted $Z(l,k)$. Describe in detail the operation of this noise reduction system and state the information needed for the system to operate correctly. [5]

c) The single-channel noise reduction system of part b) is formulated as

$$Y(l,k) = H(l,k)Z(l,k).$$

Determine the expression for $H(l,k)$. [4]

d) A multi-channel speech processing system, such as might be employed in a smartphone device for example, uses two closely spaced microphones. It can be assumed that both microphones receive the same speech signal $s(n)$. The speech signals $s_1(n)$ at microphone 1 and $s_2(n)$ at microphone 2 are corrupted by independent additive noise sources $e_1(n)$ and $e_2(n)$, respectively, so that

$$\begin{aligned} s_1(n) &= s(n) + e_1(n) \\ s_2(n) &= s(n) + e_2(n). \end{aligned}$$

The noise signals have the following properties:

$$\begin{aligned} \mathrm{E}(e_1(n)) &= \mathrm{E}(e_2(n)) = 0 \\ \mathrm{E}(e_1^2(n)) &= \sigma_1^2 \\ \mathrm{E}(e_2^2(n)) &= \sigma_2^2, \end{aligned}$$

where $\mathrm{E}(\cdot)$ denotes the expectation operator.

A multichannel noise reduction system is now implemented to give an output

$$y(n) = ax_1(n) + (1-a)x_2(n)$$

and a is a scalar constant to be determined.

i) Determine the residual noise remaining after noise reduction, $e_3(n)$, given that

$$y(n) = s(n) + e_3(n).$$

[2]

ii) Find the value of the constant a that minimizes the residual noise power of $e_3(n)$. [5]